

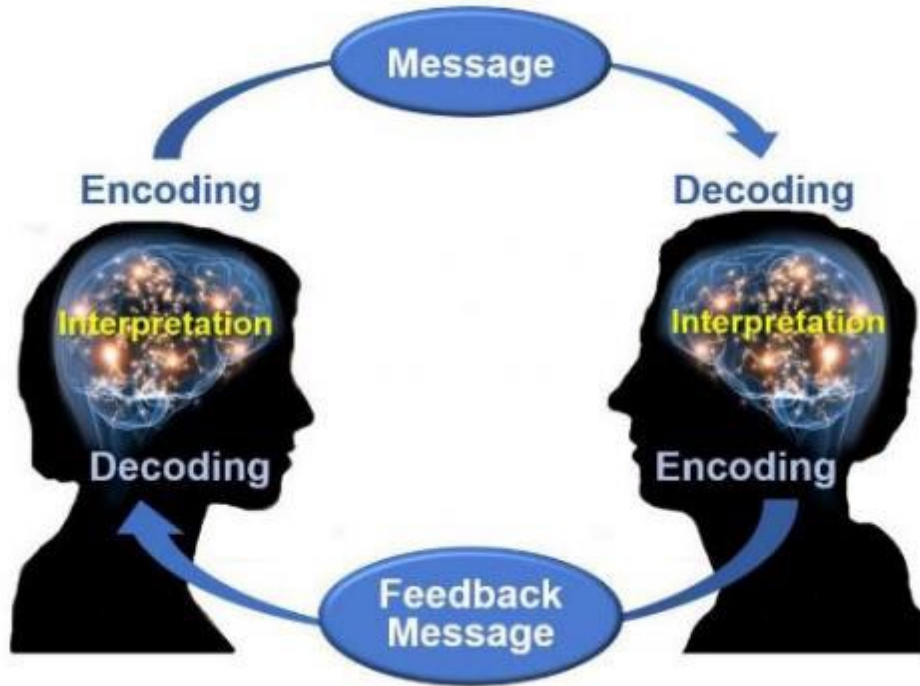
CNN for *Personalized* Speech Recognition

Dr. Grantham Pang
The University of Hong Kong

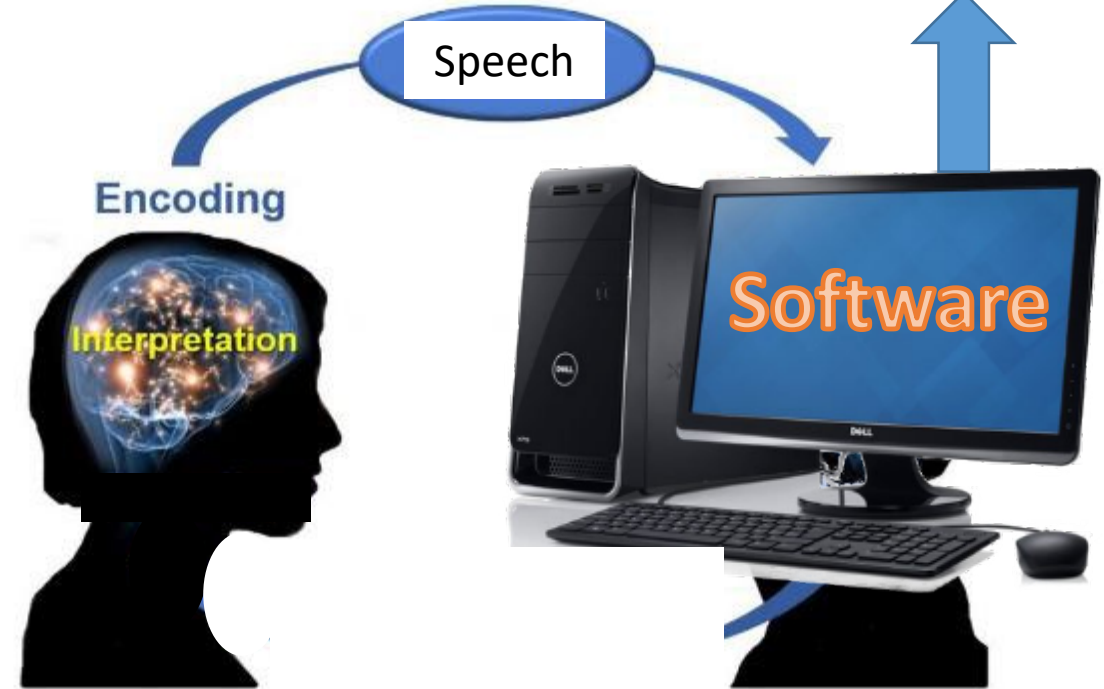
gpang@eee.hku.hk

Sept 28, 2026

Motivation



General communication



Personalized Speech Understanding

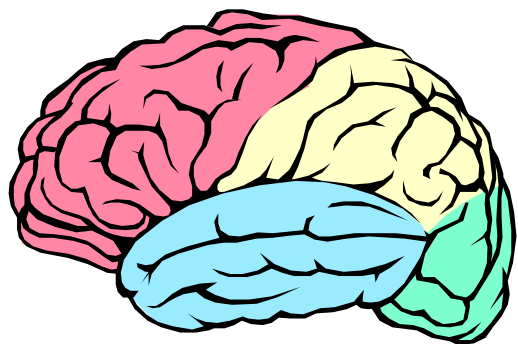
- The aim of this work is to develop personalized speech recognition software based on AI.
- Training of the software is based on audio samples collected from the person, and the trained AI model is tailored to individual.

Aphasic (失语症) speech recognition

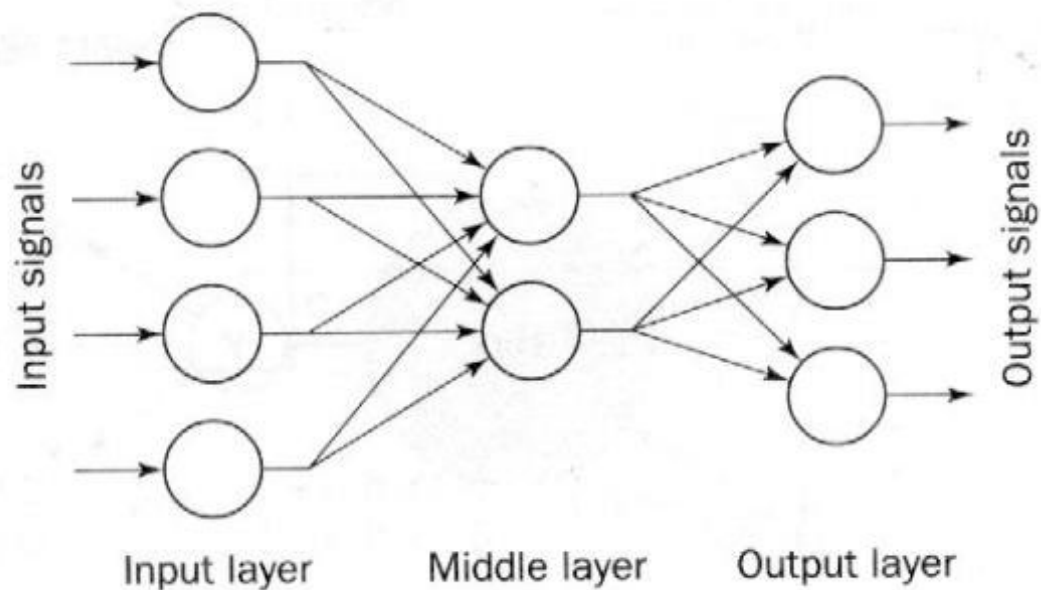
A person with **aphasia** may be unable to comprehend or unable to formulate language because of damage to specific brain regions or brain tumors. The major causes are stroke and head trauma.

A person with **aphasia** 's language must be significantly impaired in communication which would affect their daily life in different grades depending on the severity of the disorder. In this work, we explore the application of deep learning methods to perform aphasic speech recognition in Chinese. The aim is to recognize isolated words within small vocabularies for Cantonese and two dialects.

What Software ?



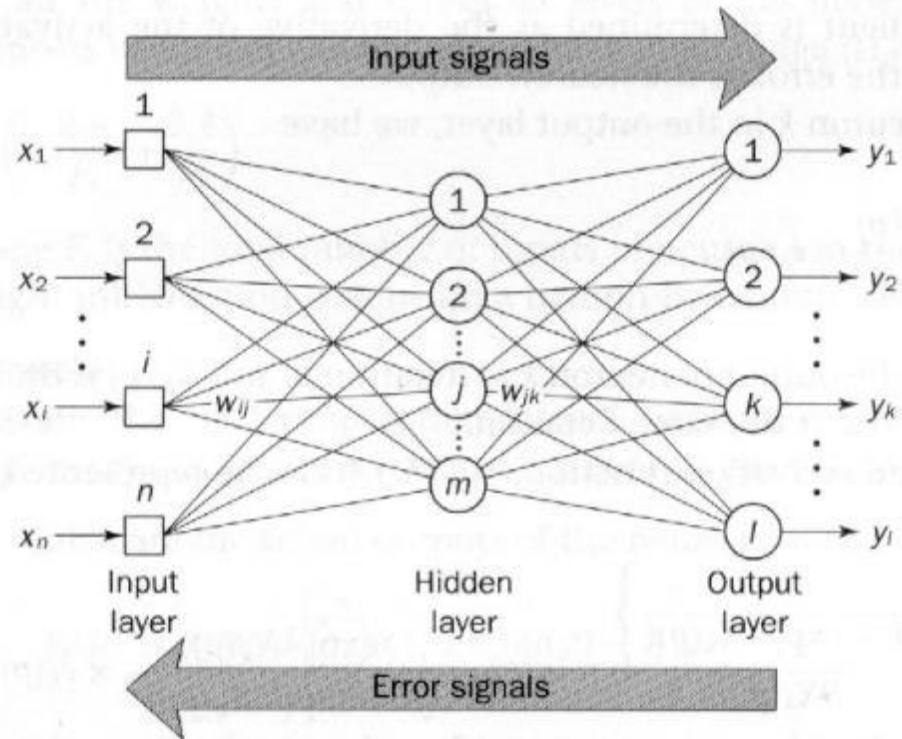
**Our brain is a
biological
neural network**



Neural Network

Why Neural Networks ?

Learning capability !

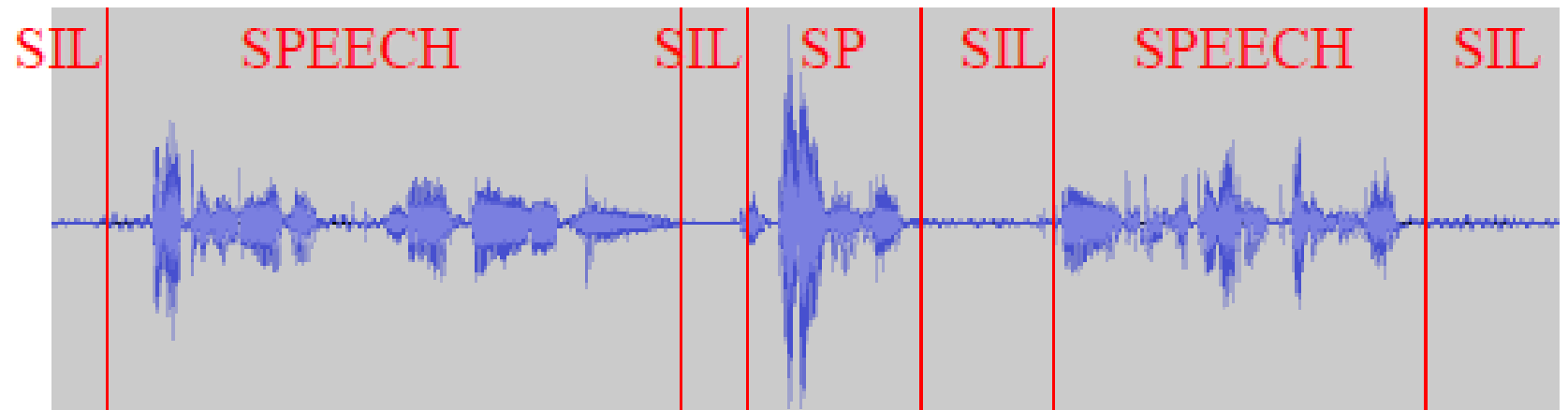


**Back-Propagation
Algorithm**

**to adjust the
interconnection weights**

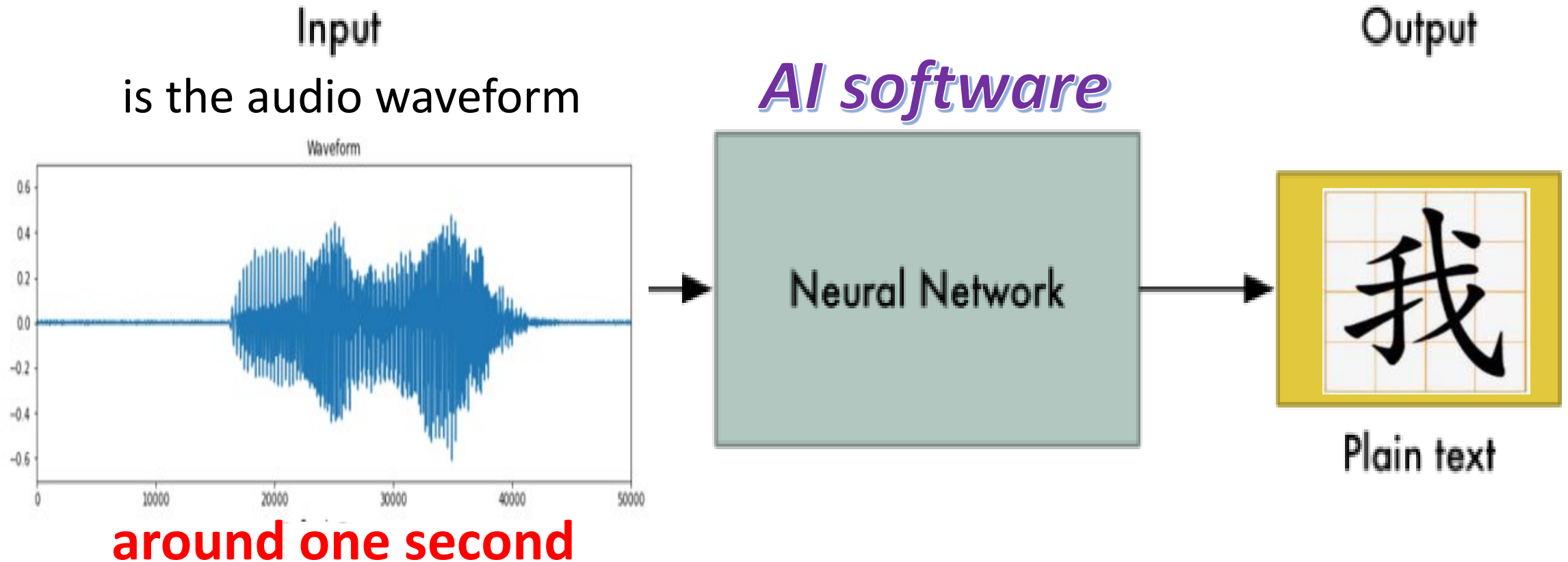
Speech Recognition

Detect periods of human speech in an audio signal



The idea:

The audio waveform of each spoken word is provided to the AI program.

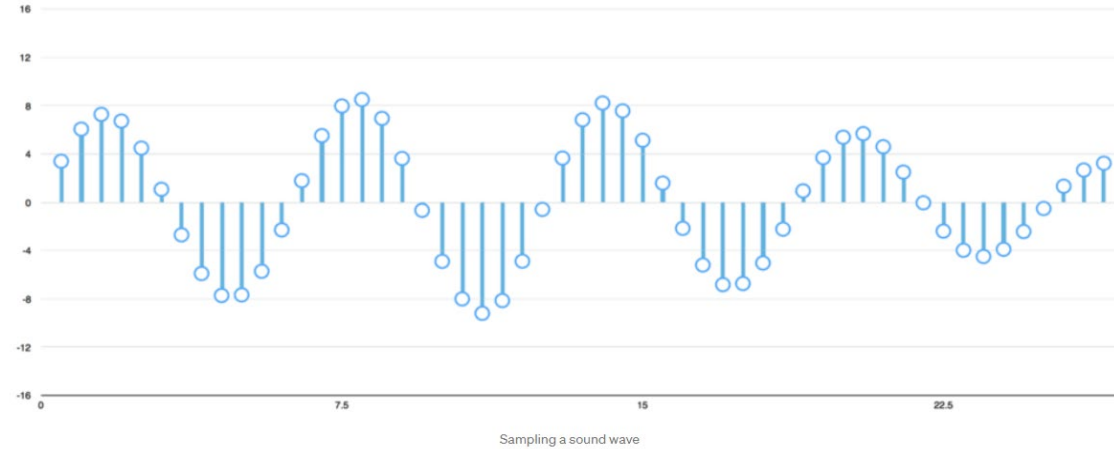


*How to convert sound waves
to suitable data for training ?*



What are spectrograms?

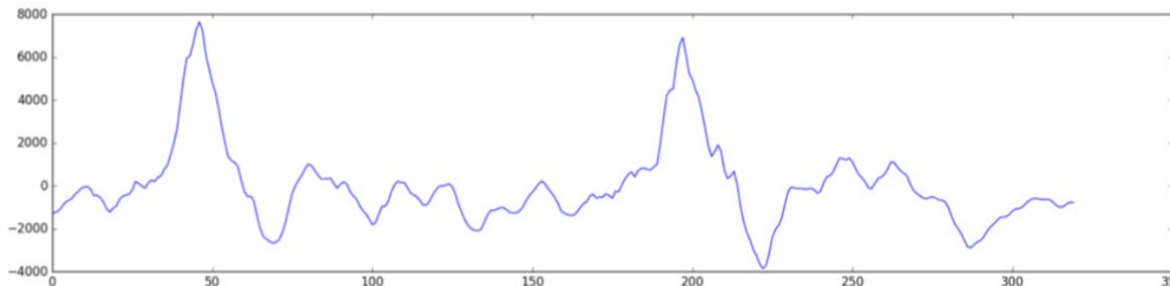
Sampling e.g. 44.1kHz or 16kHz



EXAMPLE:

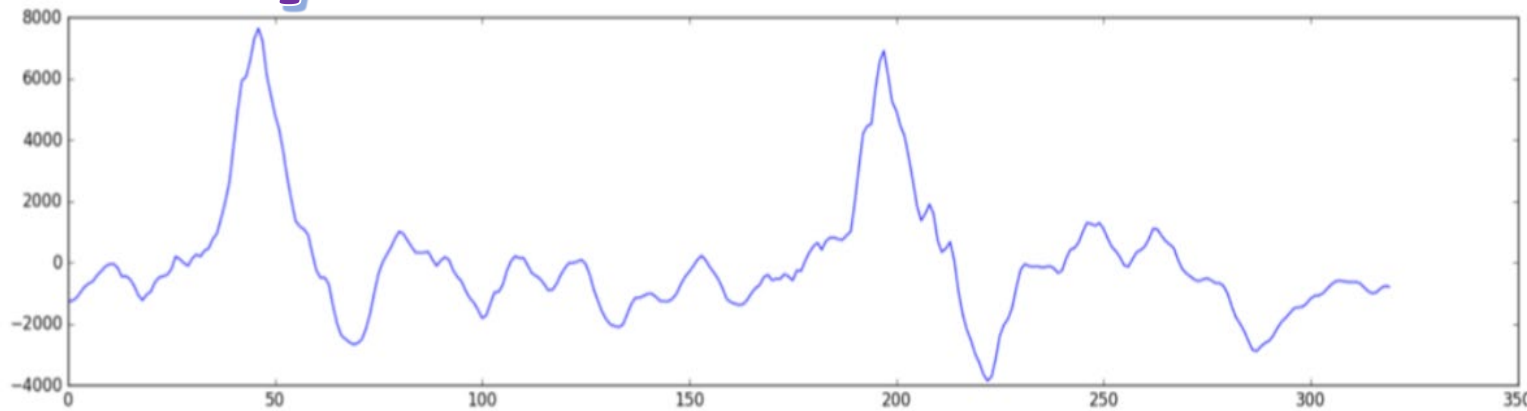
Below is a 20ms-long interval of a 16kHz signal.

Each chunk of data will have 320 samples (below)



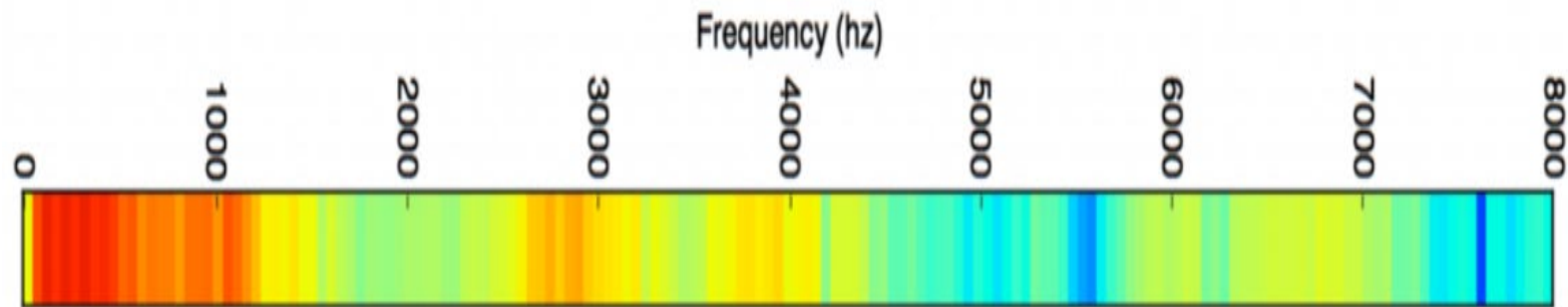
$$\begin{aligned} &16000 \times 0.02 \\ &= 320 \end{aligned}$$

Fourier Transform to show the Frequencies in the data signal



We obtain the frequency components of the 320 samples

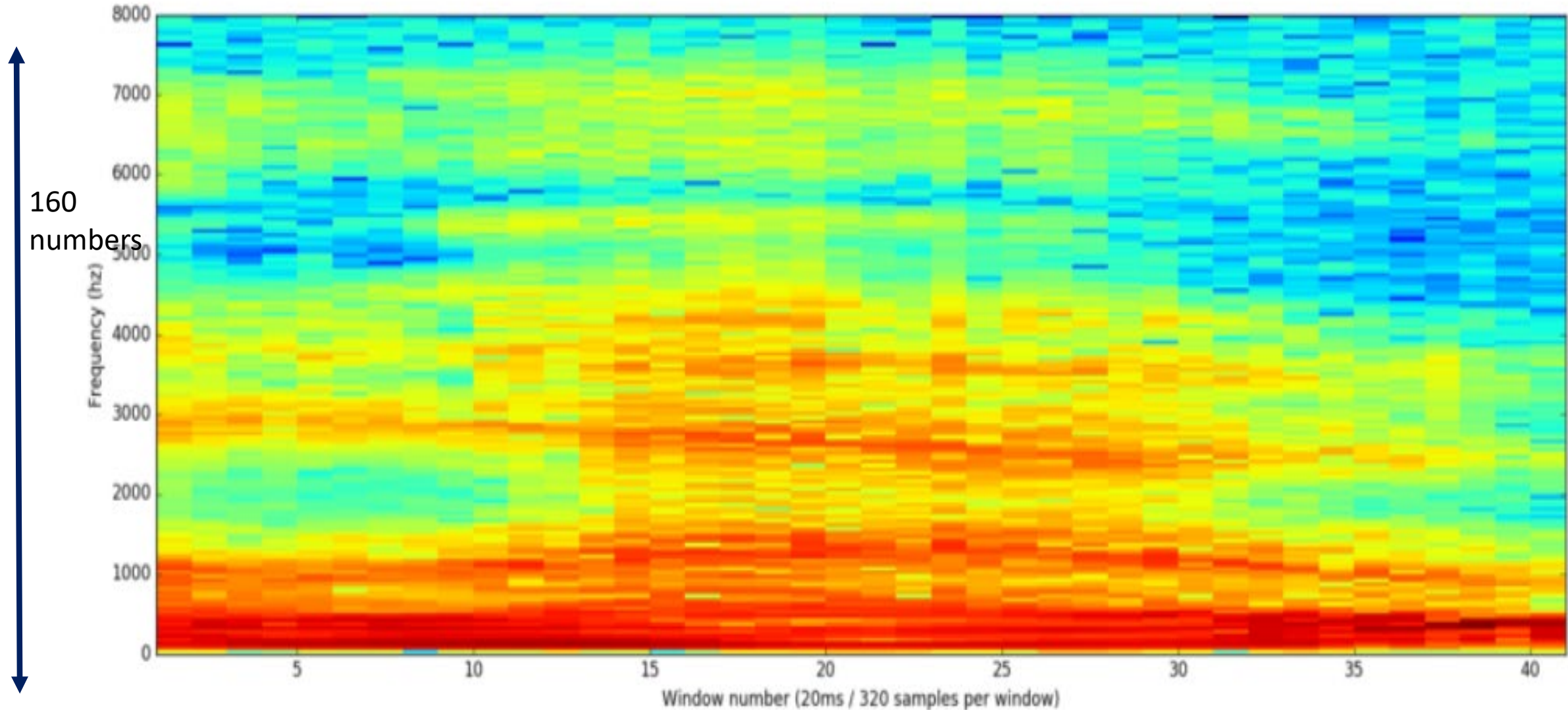
We look at audio over a frequency range up to 8000Hz. Every 50Hz is a band, and we have 160 numbers ($8000/50$) representing the amount of energy in that frequency band.



More low frequency energy with the 20ms audio.

Spectrogram

40 columns of 20ms (0.8 sec) of audio data. **Input data to the neural network.**



40 columns of 20ms (0.8 sec) of audio data.

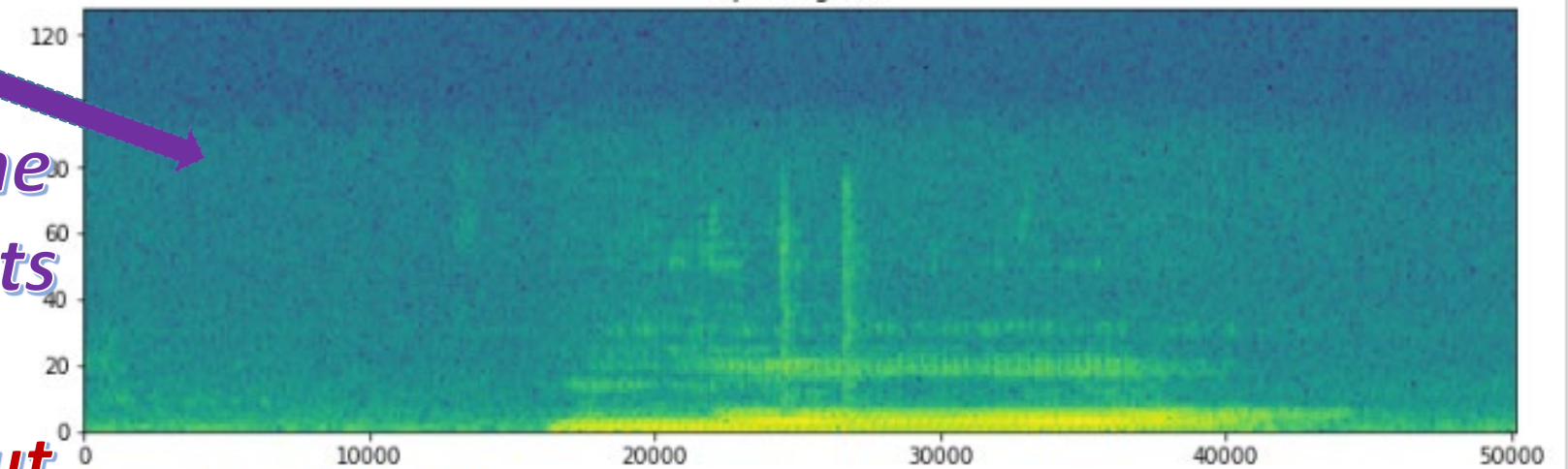
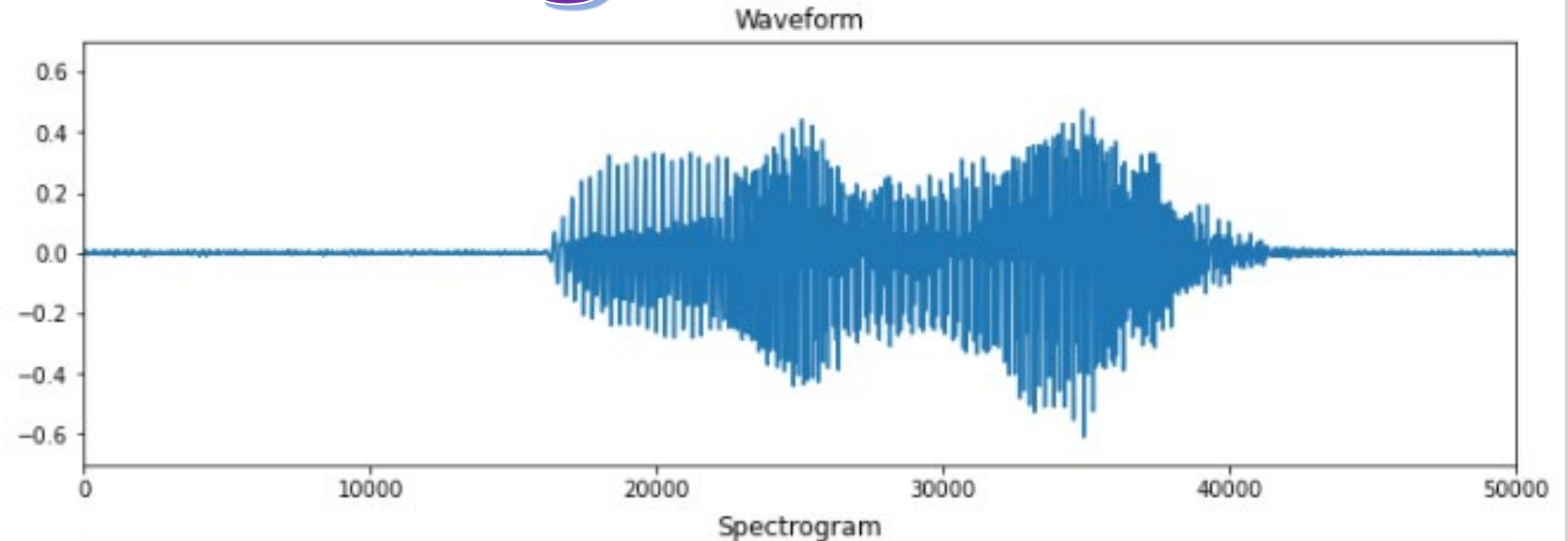
Spectrogram

The audio waveform of each spoken word cannot be provided directly as input to the AI program. The time-domain audio waveform should be **converted to the corresponding spectrogram** and used as input for deep learning.

A spectrogram is a visual representation of the **frequency content of an audio signal over time**. It is created by dividing the audio signal into short, overlapping windows and computing the Fast Fourier Transform (FFT) for each window.

This process results in a **time-frequency representation**, where the x-axis represents time, the y-axis represents frequency, and the **color intensity represents the amplitude or energy of each frequency component**.

How to do recognition ?



Spectrograms give the frequency components of the audio signal.

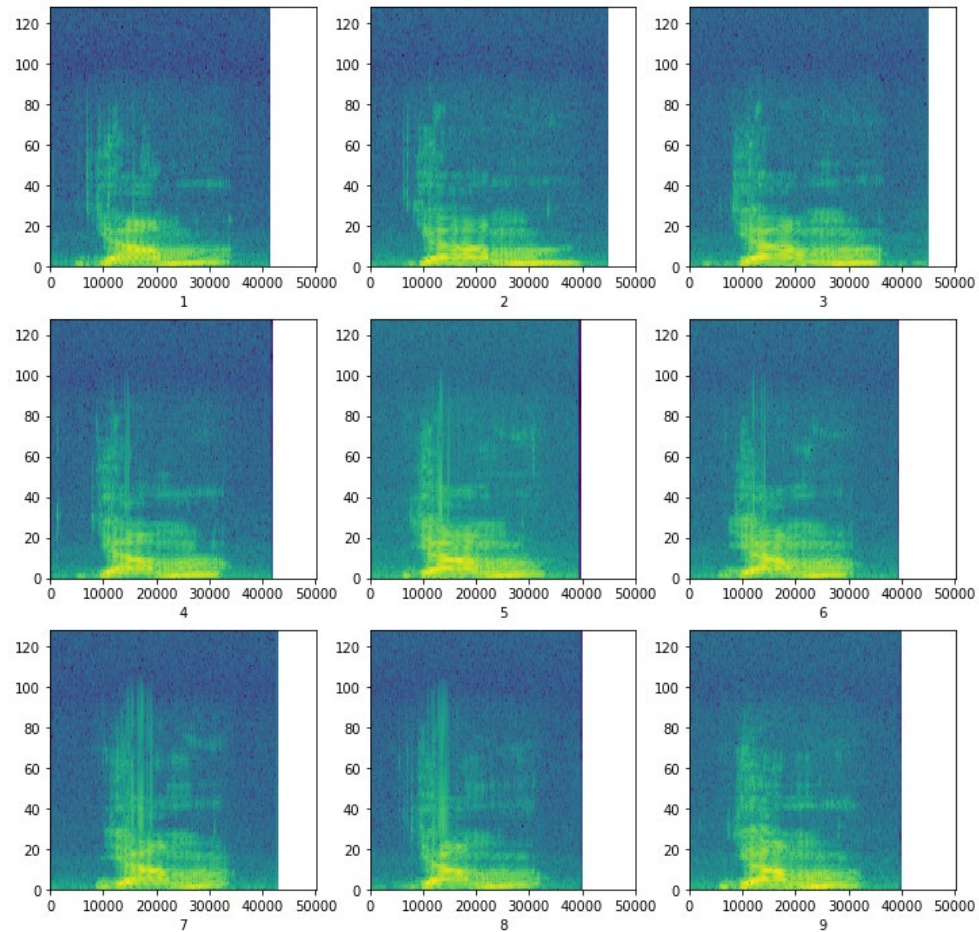
They are used as input to the neural network.

** Length of the file 50000

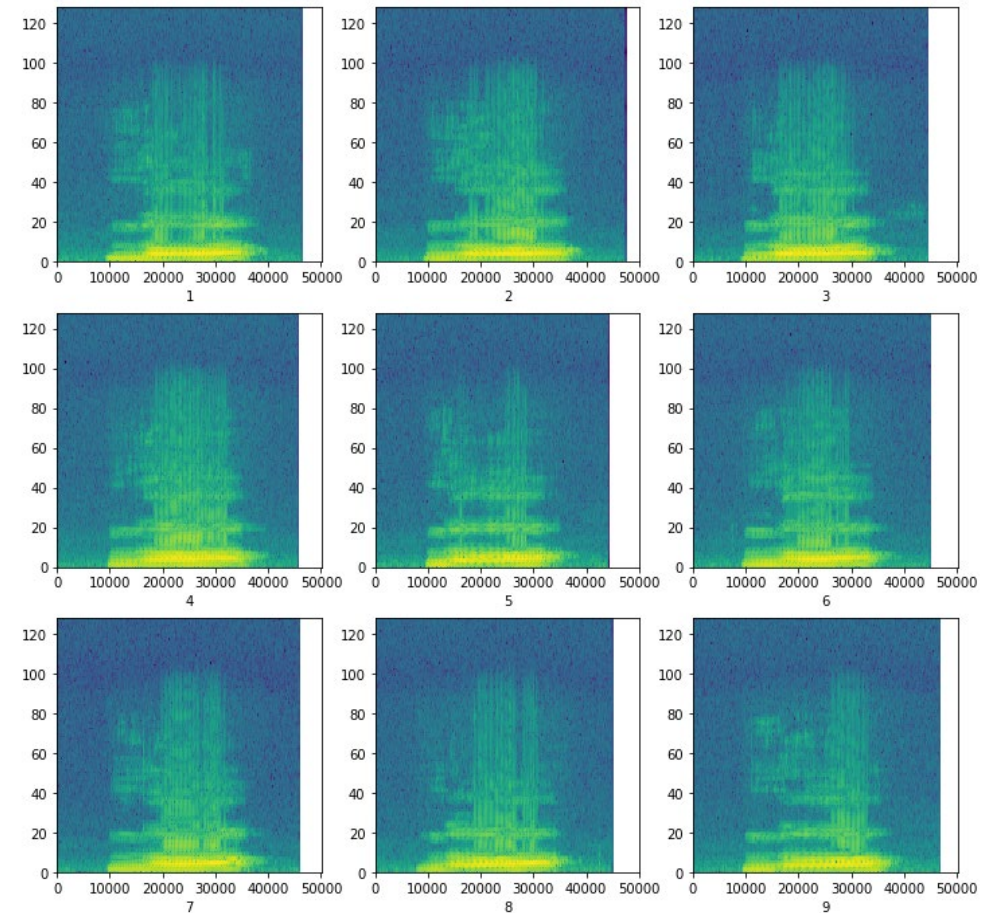
-0.002105712890625, -0.002197265625, -0.001922607421875, -0.001739501953125, -0.00183

Spectrogram shape: (389, 129)

The spectrograms of different words:



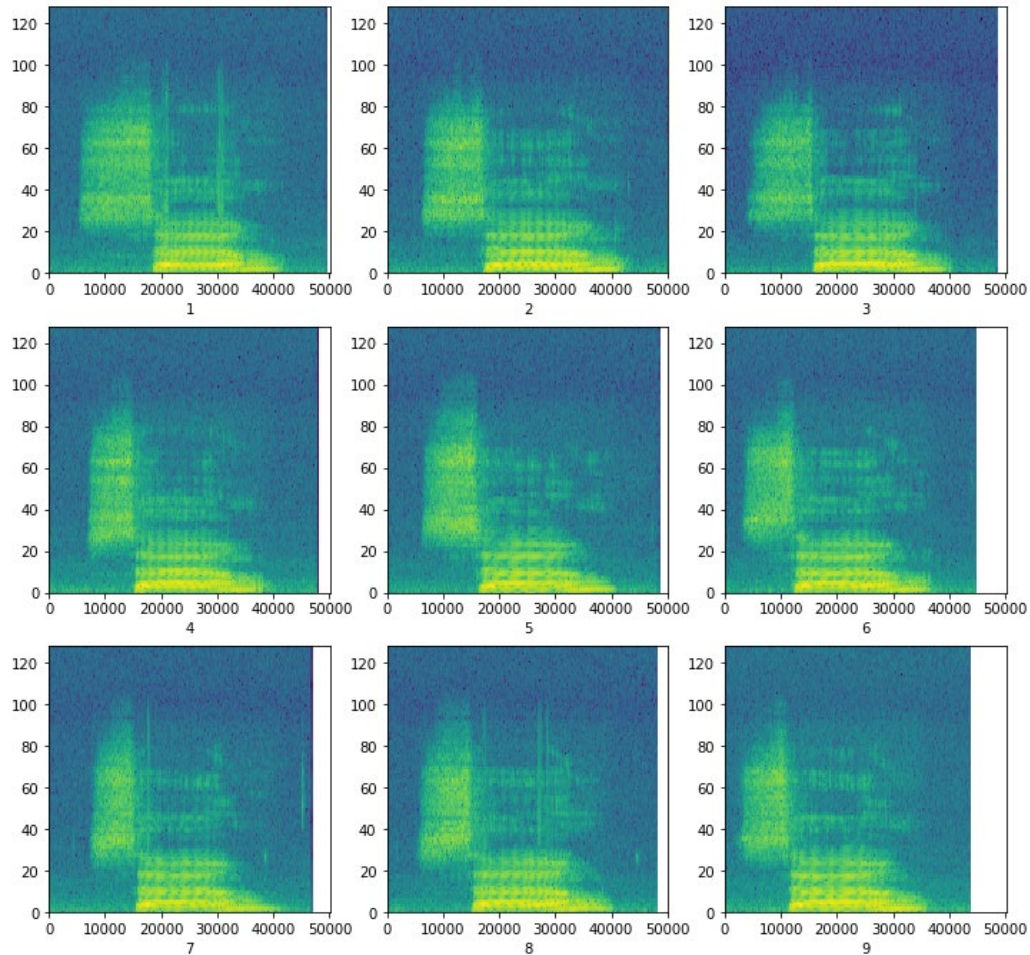
飲



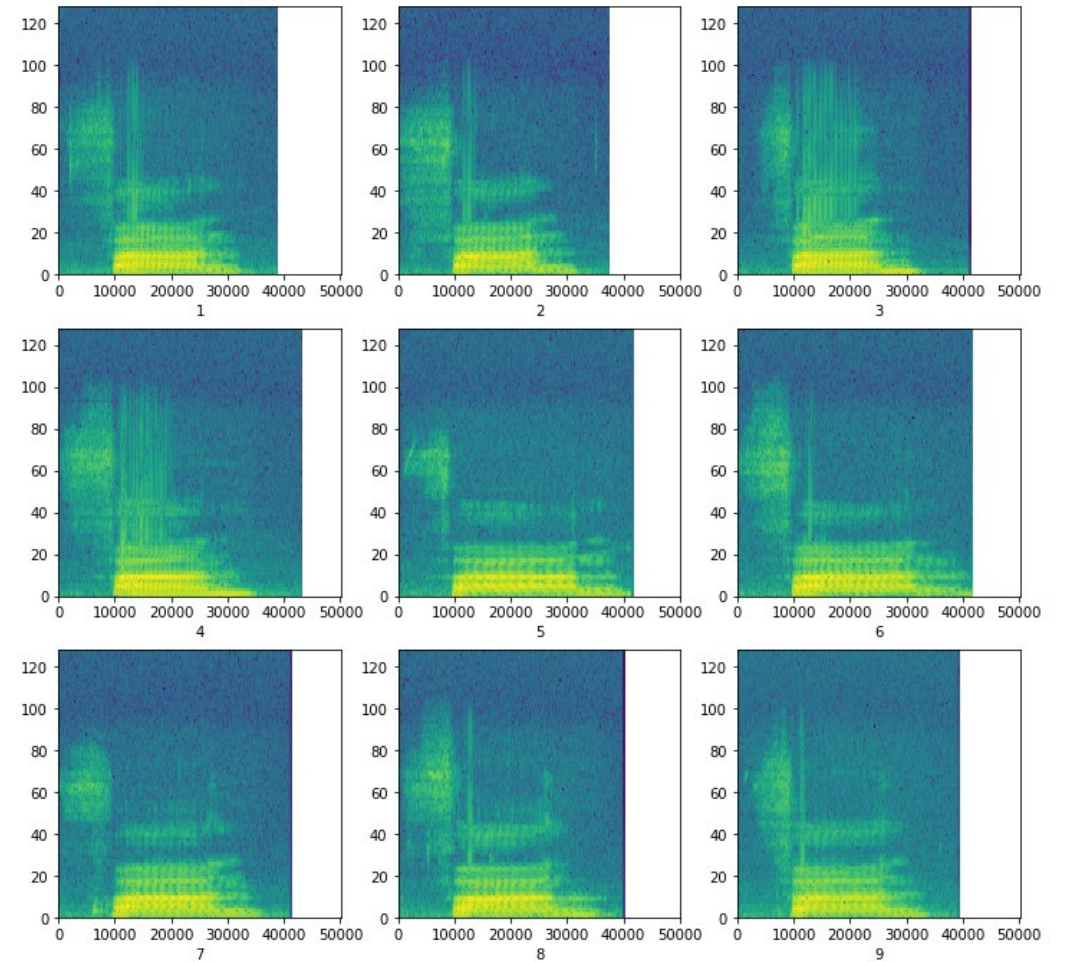
我

It can be observed that different words have different patterns of spectrogram.

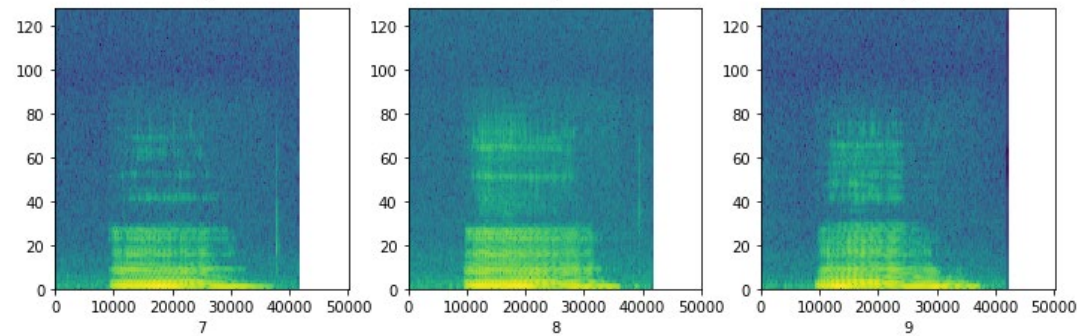
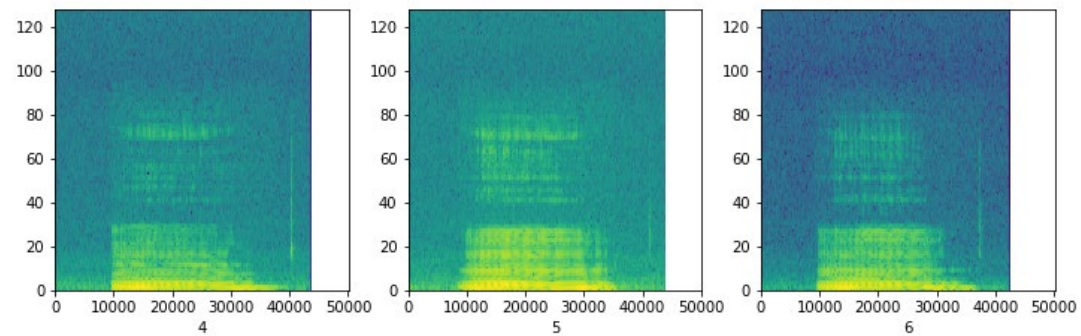
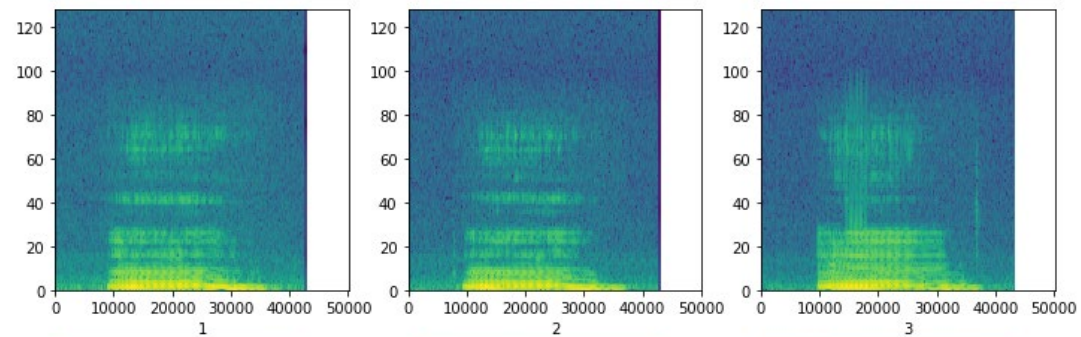
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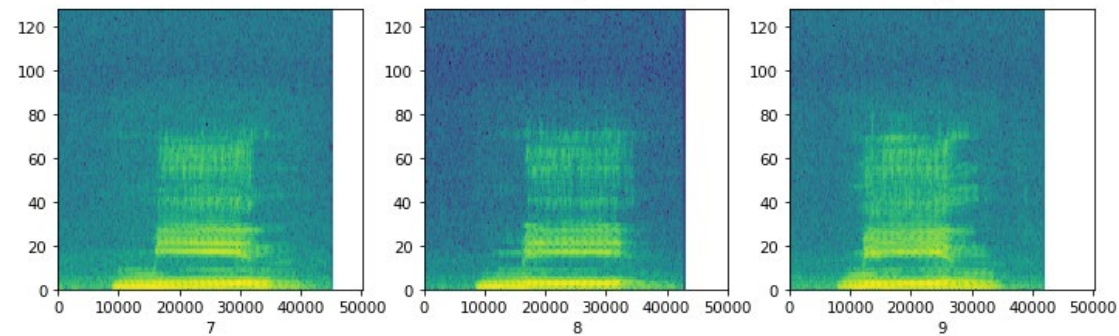
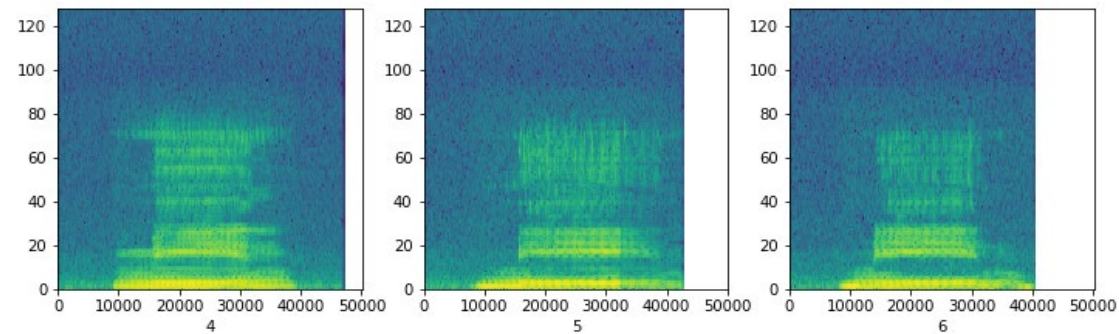
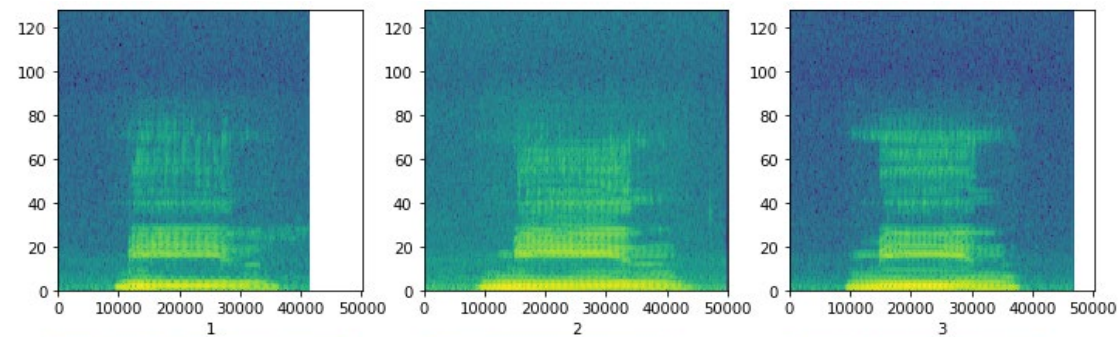
想



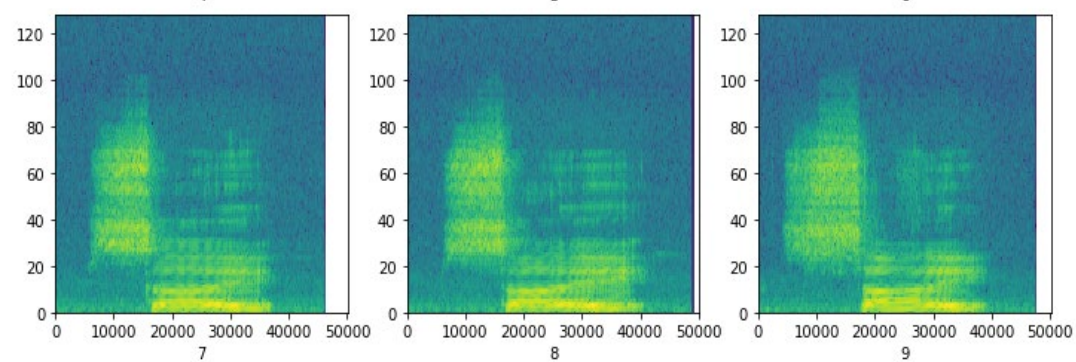
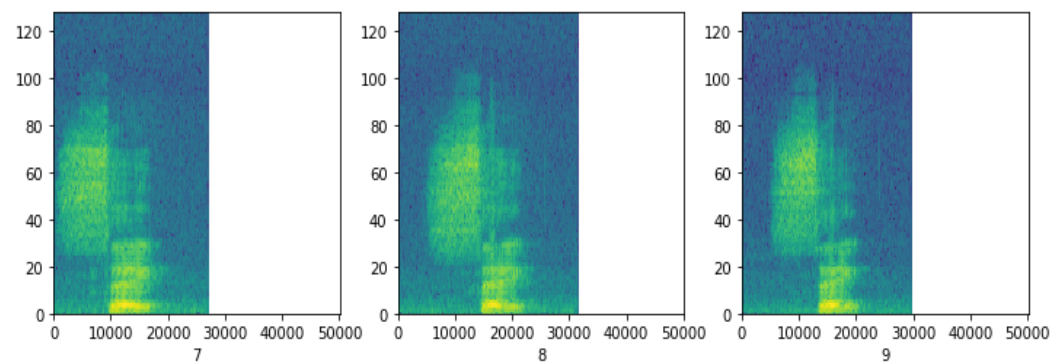
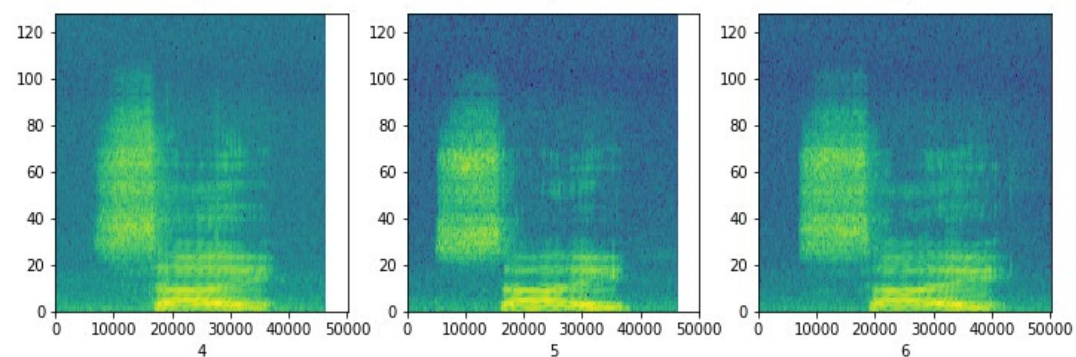
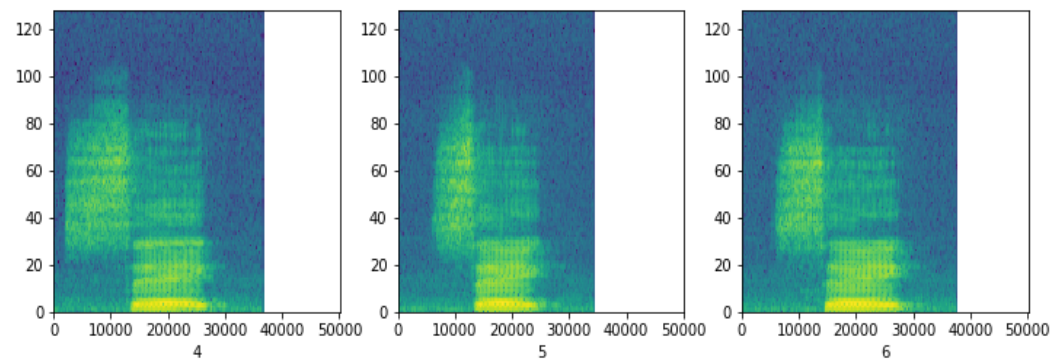
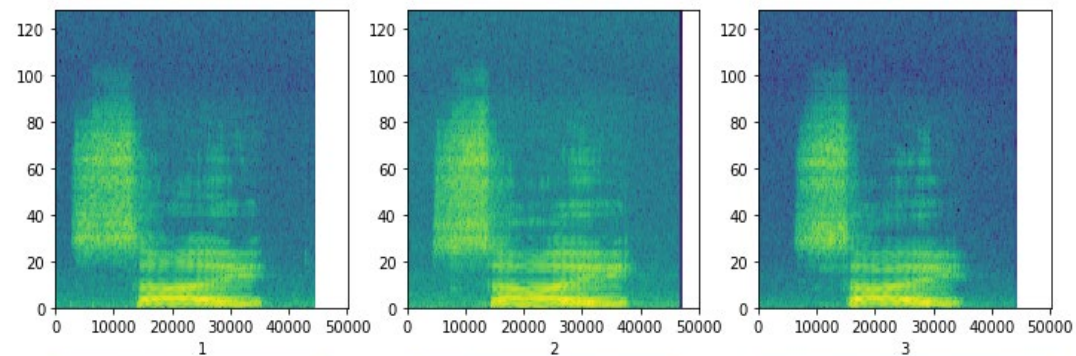
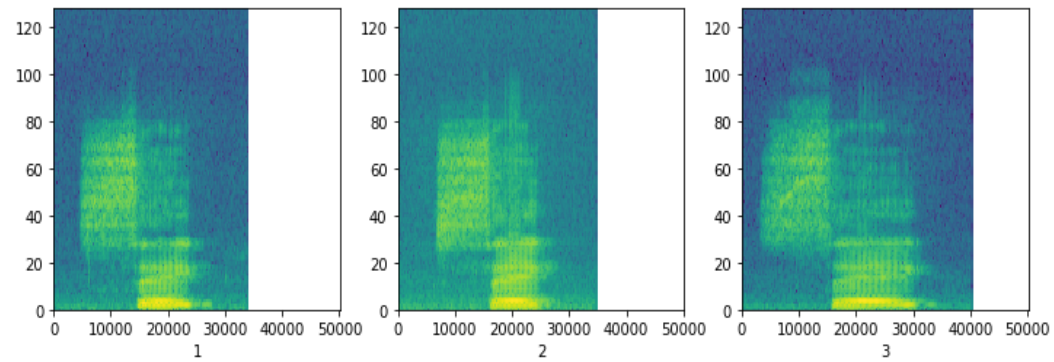
飯



唔



麵



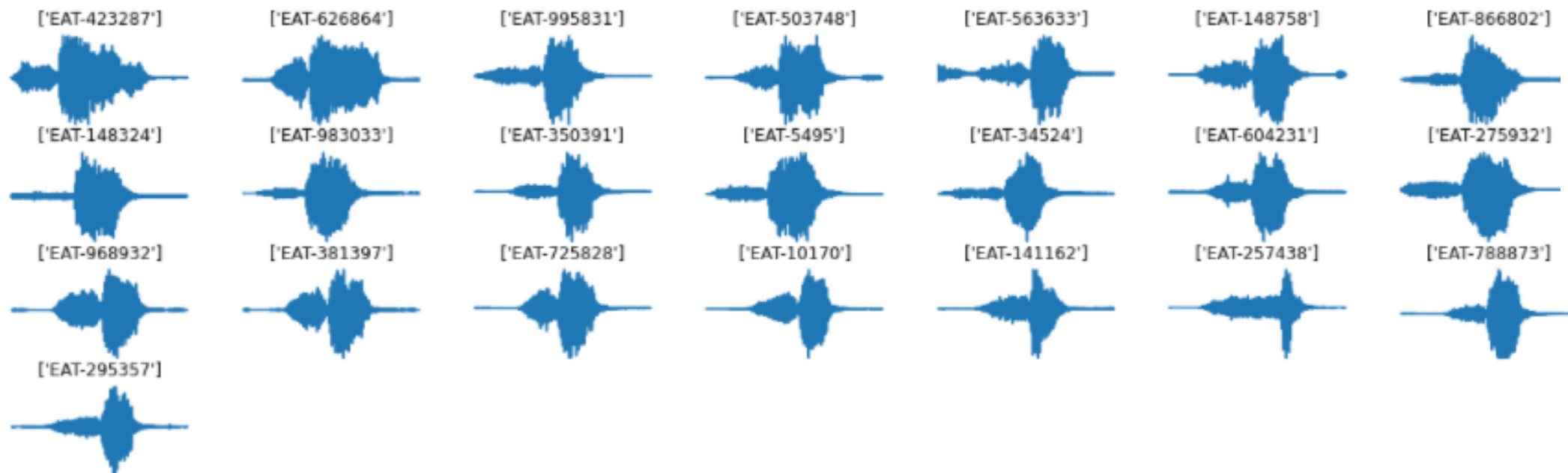
食

水

SHOWING THE AUDIO files of two words in a training database

drive/My Drive/Colab Notebooks/dataset-Moo-trainL/EAT/*

Number of files in that directory: 22



食

drive/My Drive/Colab Notebooks/dataset-Moo-trainL/RICE/*

Number of files in that directory: 10



飯

The 46 keywords using for training:

46 keywords

0	1	2	3	4	5	6	7	8	9
我	要	去	廁	所	返	睡	房	書	廚

A	B	C	D	E	F	G	H	I	J
刷	牙	洗	面	開	電	腦	門	燈	出

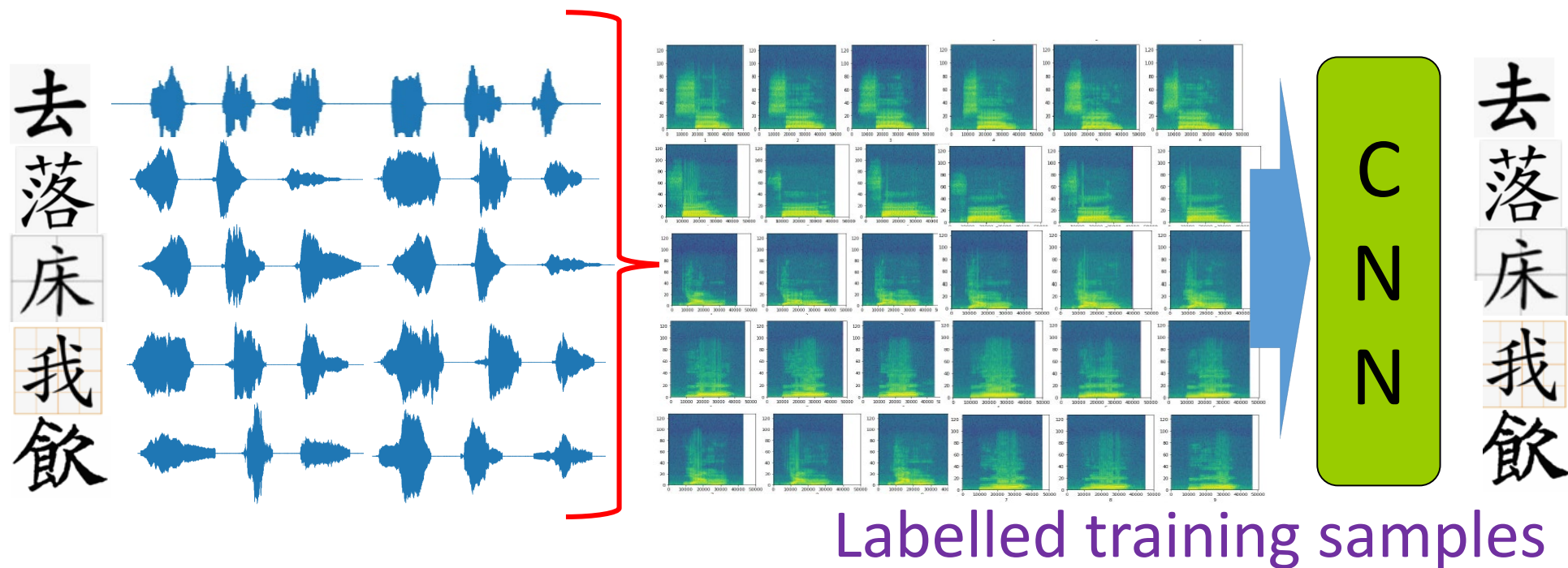
K	L	M	N	O	P	Q	R	S	T
客	廳	睇	視	叫	人	鐘	想	上	床

U	V	W	X	Y	Z
落	攤	手	機	畀	話

!	,	+	(	)	\$	%	#	@	&
你	飲	茶	唔	水	吃	麵	飯	早	餐

Training / Deep learning by CNN

Convolutional Neural Network



The difficulties of the problem:

想上

Similar sound

面麵

Same sound:
resolve by context

*Same word but different sound
according to context*

電話; 話人

*Same word but different sound
according to context*

返客廳

item: 44 ID 5ST

返上床

The dataset files (around 1GB)

	Training	Testing	
Person X	dataset-P5-20230810-org300 (300 files) dataset-P5-20230810-aug3000 (3000 files)	dataset-Test-P5-20230810-org300 (300 files)	Cantonese
Person Y	dataset-P1-20230207-aug3500 (3500 files)	dataset-Test-P1-20230207-org200 (200 files)	Shanghaiese 上海話
Person Z	dataset-P2-20230207-aug4240 (4240files)	dataset-Test-P2-20230207-org200 (200 files)	Teochew 潮洲話